THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 14: **Surface Area and Volume** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 242 cm^2	$2\pi rh$ with $r = 3.5$, $h = 11$.
2	**(d)** 671 cm^2	$2\pi r(r + h)$ with $r = 3.5$, $h = 27$.
3	**(d)** 2200 cm^2	πrl with the given values.
4	**(a)** 1617 cm^3	$\frac{1}{3}\pi r^2 h$ with the given values.
5	**(b)** 15400 cm^2	$4\pi r^2$ with $r = 35$.
6	**(d)** 130977 cm^3	$\frac{4}{3}\pi r^3$ with $r = 31.5$.
7	**(d)** 77 cm^2	$2\pi r^2$ with $r = 3.5$.
8	**(a)** $\frac{1}{3}Bh$	One third of the prism.
9	**(c)** 4752 cm^2	$2\pi rh$ with $r = 28$, $h = 27$.
10	**(a)** 1881 cm^2	$2\pi r(r + h)$ with $r = 10.5$, $h = 18$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$l = \sqrt{r^2 + h^2} = 25$ cm; CSA $= \pi rl = 550$ cm^2 (1); Volume $= \frac{1}{3}\pi r^2 h = 1232$ cm^3 (1).	1+1
12	CSA $= 2\pi r^2 = 2 \times \frac{22}{7} \times 28^2 = 4928$ cm^2 (1); TSA $= 3\pi r^2 = 7392$ cm^2 (1).	1+1
13	Base area $= 9^2 = 81$ cm^2; Volume $= \frac{1}{3} \times 81 \times 36$ (1); $= 972$ cm^3 (1).	1+1
14	$l = \sqrt{r^2 + h^2} = 70$ cm; CSA $= \pi rl = 9240$ cm^2 (1); Volume $= \frac{1}{3}\pi r^2 h = 103488$ cm^3 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$l = \sqrt{r^2 + h^2} = 50$ cm (1); CSA $= \pi rl = 2200$ cm^2 (1); Volume $= \frac{1}{3}\pi r^2 h = 9856$ cm^3 (1).	1+1+1
16	CSA $= 2\pi r^2 = 2 \times \frac{22}{7} \times 52.5^2 = 17325$ cm^2 (1); TSA $= 3\pi r^2 = 25987.5$ cm^2 (1); Volume $= \frac{2}{3}\pi r^3 = 303187.5$ cm^3 (1).	1+1+1
17	CSA $= 2\pi rh = 2 \times \frac{22}{7} \times 7 \times 35 = 1540$ cm^2 (1); Number of units of 1000 cm^2 $= \dfrac{1540}{1000} = 1.54$ (1); Cost $= 1.54 \times 10 =$ ₹15.4 (1).	1+1+1
	OR: $l = \sqrt{r^2 + h^2} = 17.5$ cm (1); CSA $= \pi rl = 577.5$ cm^2 (1); Volume $= \frac{1}{3}\pi r^2 h = 1617$ cm^3 (1).	1+1+1
18	CSA $= 2\pi r^2 = 2 \times \frac{22}{7} \times 63^2 = 24948$ cm^2 (1); TSA $= 3\pi r^2 = 37422$ cm^2 (1); Volume $= \frac{2}{3}\pi r^3 = 523908$ cm^3 (1).	1+1+1
	OR: CSA $= 2\pi rh = 2 \times \frac{22}{7} \times 14 \times 25 = 2200$ cm^2 (1); Number of units of 1000 cm^2 $= \dfrac{2200}{1000} = 2.2$ (1); Cost $= 2.2 \times 10 =$ ₹22 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$2\pi r = 44$, so $r = \dfrac{44 \times 7}{2 \times 22} = 7$ cm (1); CSA = area of the sheet $= 44 \times 25 = 1100$ cm^2 (2); Volume $= \pi r^2 h = \frac{22}{7} \times 7^2 \times 25 = 3850$ cm^3 (2).	1+2+2
	OR: Volume is conserved: $n \times \frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3$ (1); $n = \left(\dfrac{8}{2}\right)^3 = 64$ (1); Total area of small spheres $= 64 \times 4\pi(2)^2$, area of big sphere $= 4\pi(8)^2$ (1); Ratio $= \dfrac{64 \times 2^2}{8^2} = 4$: the small spheres have 4 times the surface area (2).	1+1+1+2
20	$2\pi r = 44$, so $r = \dfrac{44 \times 7}{2 \times 22} = 7$ cm (1); CSA = area of the sheet $= 44 \times 20 = 880$ cm^2 (2); Volume $= \pi r^2 h = \frac{22}{7} \times 7^2 \times 20 = 3080$ cm^3 (2).	1+2+2
	OR: Volume is conserved: $n \times \frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3$ (1); $n = \left(\dfrac{9}{3}\right)^3 = 27$ (1); Total area of small spheres $= 27 \times 4\pi(3)^2$, area of big sphere $= 4\pi(9)^2$ (1); Ratio $= \dfrac{27 \times 3^2}{9^2} = 3$: the small spheres have 3 times the surface area (2).	1+1+1+2

Mistakes to avoid:

- Forgetting the factor $\frac{1}{3}$ in the volume of a cone or pyramid.
- Using the vertical height h instead of the slant height l in $\pi r l$ (CSA of a cone).
- Writing the TSA of a cylinder as $2\pi rh + \pi r^2$; there are two circular ends, so it is $2\pi rh + 2\pi r^2$.
- Using the diameter as the radius, or leaving r in the formula when the problem gives d.
- Mixing units: converting cm^3 to litres needs $\div 1000$, and m^3 to litres needs $\times 1000$.
- For a hemisphere, using the sphere formulas without halving the volume, or taking TSA as $2\pi r^2$ and forgetting the flat circle ($3\pi r^2$ in all).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 14: **Surface Area and Volume** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 550 cm^2	πrl with the given values.
2	**(b)** 1232 cm^3	$\frac{1}{3}\pi r^2 h$ with the given values.
3	**(a)** 1386 cm^2	$4\pi r^2$ with $r = 10.5$.
4	**(c)** 4851 cm^3	$\frac{4}{3}\pi r^3$ with $r = 10.5$.
5	**(a)** 1232 cm^2	$2\pi r^2$ with $r = 14$.
6	**(c)** $2\pi r(r + h)$	Curved surface plus two circular ends.
7	**(a)** 198 cm^2	$2\pi rh$ with $r = 3.5$, $h = 9$.
8	**(c)** 715 cm^2	$2\pi r(r + h)$ with $r = 3.5$, $h = 29$.
9	**(c)** 2502.5 cm^2	πrl with the given values.
10	**(a)** 103488 cm^3	$\frac{1}{3}\pi r^2 h$ with the given values.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	CSA $= 2\pi r^2 = 2 \times \frac{22}{7} \times 14^2 = 1232$ cm^2 (1); TSA $= 3\pi r^2 = 1848$ cm^2 (1).	1+1
12	Base area $= 3^2 = 9$ cm^2; Volume $= \frac{1}{3} \times 9 \times 36$ (1); $= 108$ cm^3 (1).	1+1
13	$l = \sqrt{r^2 + h^2} = 59.5$ cm; CSA $= \pi rl = 5236$ cm^2 (1); Volume $= \frac{1}{3}\pi r^2 h = 43120$ cm^3 (1).	1+1
14	CSA $= 2\pi r^2 = 2 \times \frac{22}{7} \times 7^2 = 308$ cm^2 (1); TSA $= 3\pi r^2 = 462$ cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$l = \sqrt{r^2 + h^2} = 70$ cm (1); CSA $= \pi rl = 9240$ cm^2 (1); Volume $= \frac{1}{3}\pi r^2 h = 103488$ cm^3 (1).	1+1+1
16	CSA $= 2\pi r^2 = 2 \times \frac{22}{7} \times 31.5^2 = 6237$ cm^2 (1); TSA $= 3\pi r^2 = 9355.5$ cm^2 (1); Volume $= \frac{2}{3}\pi r^3 = 65488.5$ cm^3 (1).	1+1+1
17	CSA $= 2\pi rh = 2 \times \frac{22}{7} \times 21 \times 10 = 1320$ cm^2 (1); Number of units of 1000 cm^2 $= \frac{1320}{1000} = 1.32$ (1); Cost $= 1.32 \times 30 =$ ₹39.6 (1).	1+1+1
	OR: $l = \sqrt{r^2 + h^2} = 35$ cm (1); CSA $= \pi rl = 2310$ cm^2 (1); Volume $= \frac{1}{3}\pi r^2 h = 12936$ cm^3 (1).	1+1+1
18	CSA $= 2\pi r^2 = 2 \times \frac{22}{7} \times 42^2 = 11088$ cm^2 (1); TSA $= 3\pi r^2 = 16632$ cm^2 (1); Volume $= \frac{2}{3}\pi r^3 = 155232$ cm^3 (1).	1+1+1
	OR: CSA $= 2\pi rh = 2 \times \frac{22}{7} \times 14 \times 30 = 2640$ cm^2 (1); Number of units of 1000 cm^2 $= \frac{2640}{1000} = 2.64$ (1); Cost $= 2.64 \times 35 =$ ₹92.4 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$2\pi r = 88$, so $r = \frac{88 \times 7}{2 \times 22} = 14$ cm (1); CSA = area of the sheet $= 88 \times 5 = 440$ cm^2 (2); Volume $= \pi r^2 h = \frac{22}{7} \times 14^2 \times 5 = 3080$ cm^3 (2).	1+2+2
	OR: Volume is conserved: $n \times \frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3$ (1); $n = \left(\frac{3}{1}\right)^3 = 27$ (1); Total area of small spheres $= 27 \times 4\pi(1)^2$, area of big sphere $= 4\pi(3)^2$ (1); Ratio $= \frac{27 \times 1^2}{3^2} = 3$: the small spheres have 3 times the surface area (2).	1+1+1+2
20	$2\pi r = 88$, so $r = \frac{88 \times 7}{2 \times 22} = 14$ cm (1); CSA = area of the sheet $= 88 \times 8 = 704$ cm^2 (2); Volume $= \pi r^2 h = \frac{22}{7} \times 14^2 \times 8 = 4928$ cm^3 (2).	1+2+2
	OR: Volume is conserved: $n \times \frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3$ (1); $n = \left(\frac{12}{3}\right)^3 = 64$ (1); Total area of small spheres $= 64 \times 4\pi(3)^2$, area of big sphere $= 4\pi(12)^2$ (1); Ratio $= \frac{64 \times 3^2}{12^2} = 4$: the small spheres have 4 times the surface area (2).	1+1+1+2

Mistakes to avoid:

- Forgetting the factor $\frac{1}{3}$ in the volume of a cone or pyramid.
- Using the vertical height h instead of the slant height l in $\pi r l$ (CSA of a cone).
- Writing the TSA of a cylinder as $2\pi rh + \pi r^2$; there are two circular ends, so it is $2\pi rh + 2\pi r^2$.
- Using the diameter as the radius, or leaving r in the formula when the problem gives d.
- Mixing units: converting cm^3 to litres needs $\div 1000$, and m^3 to litres needs $\times 1000$.
- For a hemisphere, using the sphere formulas without halving the volume, or taking TSA as $2\pi r^2$ and forgetting the flat circle ($3\pi r^2$ in all).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 14: **Surface Area and Volume** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 5544 cm^2	$4\pi r^2$ with $r = 21$.
2	**(a)** 38808 cm^3	$\frac{4}{3}\pi r^3$ with $r = 21$.
3	**(c)** 308 cm^2	$2\pi r^2$ with $r = 7$.
4	**(c)** cubic units	Three dimensions.
5	**(d)** 3960 cm^2	$2\pi rh$ with $r = 21$, $h = 30$.
6	**(b)** 1683 cm^2	$2\pi r(r + h)$ with $r = 10.5$, $h = 15$.
7	**(a)** 9240 cm^2	πrl with the given values.
8	**(b)** 13475 cm^3	$\frac{1}{3}\pi r^2 h$ with the given values.
9	**(d)** 9856 cm^2	$4\pi r^2$ with $r = 28$.
10	**(d)** 310464 cm^3	$\frac{4}{3}\pi r^3$ with $r = 42$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Base area $= 3^2 = 9$ cm^2; Volume $= \frac{1}{3} \times 9 \times 21$ (1); $= 63$ cm^3 (1).	1+1
12	$l = \sqrt{r^2 + h^2} = 45.5$ cm; CSA $= \pi rl = 2502.5$ cm^2 (1); Volume $= \frac{1}{3}\pi r^2 h = 13475$ cm^3 (1).	1+1
13	CSA $= 2\pi r^2 = 2 \times \frac{22}{7} \times 3.5^2 = 77$ cm^2 (1); TSA $= 3\pi r^2 = 115.5$ cm^2 (1).	1+1
14	Base area $= 11^2 = 121$ cm^2; Volume $= \frac{1}{3} \times 121 \times 15$ (1); $= 605$ cm^3 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$l = \sqrt{r^2 + h^2} = 45.5$ cm (1); CSA $= \pi rl = 2502.5$ cm^2 (1); Volume $= \frac{1}{3}\pi r^2 h = 13475$ cm^3 (1).	1+1+1
16	CSA $= 2\pi r^2 = 2 \times \frac{22}{7} \times 21^2 = 2772$ cm^2 (1); TSA $= 3\pi r^2 = 4158$ cm^2 (1); Volume $= \frac{2}{3}\pi r^3 = 19404$ cm^3 (1).	1+1+1
17	CSA $= 2\pi rh = 2 \times \frac{22}{7} \times 7 \times 15 = 660$ cm^2 (1); Number of units of 1000 cm^2 $= \frac{660}{1000} = 0.66$ (1); Cost $= 0.66 \times 30 =$ ₹19.8 (1).	1+1+1
	OR: $l = \sqrt{r^2 + h^2} = 25$ cm (1); CSA $= \pi rl = 550$ cm^2 (1); Volume $= \frac{1}{3}\pi r^2 h = 1232$ cm^3 (1).	1+1+1
18	CSA $= 2\pi r^2 = 2 \times \frac{22}{7} \times 10.5^2 = 693$ cm^2 (1); TSA $= 3\pi r^2 = 1039.5$ cm^2 (1); Volume $= \frac{2}{3}\pi r^3 = 2425.5$ cm^3 (1).	1+1+1
	OR: CSA $= 2\pi rh = 2 \times \frac{22}{7} \times 21 \times 20 = 2640$ cm^2 (1); Number of units of 1000 cm^2 $= \frac{2640}{1000} = 2.64$ (1); Cost $= 2.64 \times 20 =$ ₹52.8 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$2\pi r = 132$, so $r = \frac{132 \times 7}{2 \times 22} = 21$ cm (1); CSA = area of the sheet $= 132 \times 5 = 660$ cm^2 (2); Volume $= \pi r^2 h = \frac{22}{7} \times 21^2 \times 5 = 6930$ cm^3 (2).	1+2+2
	OR: Volume is conserved: $n \times \frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3$ (1); $n = \left(\frac{25}{5}\right)^3 = 125$ (1); Total area of small spheres $= 125 \times 4\pi(5)^2$, area of big sphere $= 4\pi(25)^2$ (1); Ratio $= \frac{125 \times 5^2}{25^2} = 5$: the small spheres have 5 times the surface area (2).	1+1+1+2
20	$2\pi r = 88$, so $r = \frac{88 \times 7}{2 \times 22} = 14$ cm (1); CSA = area of the sheet $= 88 \times 16 = 1408$ cm^2 (2); Volume $= \pi r^2 h = \frac{22}{7} \times 14^2 \times 16 = 9856$ cm^3 (2).	1+2+2
	OR: Volume is conserved: $n \times \frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3$ (1); $n = \left(\frac{30}{5}\right)^3 = 216$ (1); Total area of small spheres $= 216 \times 4\pi(5)^2$, area of big sphere $= 4\pi(30)^2$ (1); Ratio $= \frac{216 \times 5^2}{30^2} = 6$: the small spheres have 6 times the surface area (2).	1+1+1+2

Mistakes to avoid:

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- Using the vertical height h instead of the slant height l in $\pi r l$ (CSA of a cone).
- Writing the TSA of a cylinder as $2\pi rh + \pi r^2$; there are two circular ends, so it is $2\pi rh + 2\pi r^2$.
- Using the diameter as the radius, or leaving r in the formula when the problem gives d.
- Mixing units: converting cm^3 to litres needs $\div 1000$, and m^3 to litres needs $\times 1000$.
- For a hemisphere, using the sphere formulas without halving the volume, or taking TSA as $2\pi r^2$ and forgetting the flat circle ($3\pi r^2$ in all).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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